

# GCSE algebra - Set 2 - Solutions

Specification M4.1-M4.8, M4.18, M4.19.

## Answer key

| Question | Answer | Topic        |
|----------|--------|--------------|
| 1        | C      | GCSE algebra |
| 2        | A      | GCSE algebra |
| 3        | E      | GCSE algebra |

## Worked solutions

### Question 1 ALG-15 · GCSE algebra · spec M4.18

The distance-time graph of a journey is a straight line from (0, 0) to (2, 60), then horizontal to (3, 60), then a straight line to (5, 0) (time in hours, distance from home in km). What was the average speed for the whole journey?

**Answer: C**

Total distance  $60 + 60 = 120$  km in 5 hours: 24 km/h. (Displacement is 0 but distance travelled is not.)

#### Full method - how and why

Average speed uses **total distance**, not net displacement - and this journey returns home.

- 1 Read the outward distance: the traveller goes from 0 to 60 km, so 60 km out.  
*why:* The graph shows distance **from home** against time.
- 2 The horizontal section is a stop (no distance covered), and the final section returns to 0: another 60 km.  
*why:* Coming back still counts as distance travelled.
- 3 Total distance  $= 60 + 60 = 120$  km over a total time of 5 hours.  
*why:* Include the stopped hour in the time - it is part of the journey.
- 4 Average speed  $= \frac{120}{5} = 24$  km/h.  
*why:* Total distance divided by total time.

#### Where it goes wrong

- Using the net displacement 0 and answering 0 km/h - an offered option.
- Excluding the stopped hour, giving 30 km/h.

**Takeaway** Average speed = total distance  $\div$  total time; stops count in the time, return legs count in the distance.

### Question 2 ALG-03 · GCSE algebra · spec M4.6

Simplify  $\frac{x^2 - 9}{x^2 + x - 12} \times \frac{x^2 - 16}{x^2 - 3x}$ .

**Answer: A**

Factorise everything:  $\frac{(x-3)(x+3)}{(x+3)(x-4)} \times \frac{(x-4)(x+4)}{x(x-3)}$ . The  $(x+4)$  factors cancel and one  $(x-3)$  cancels, leaving  $\frac{x+3}{x(x-3)}$ .

#### Full method - how and why

Multiplying algebraic fractions: factorise everything first, then cancel common factors across the product.

- 1 Factorise each part:  $x^2 - 9 = (x-3)(x+3)$ ;  $x^2 + x - 12 = (x+4)(x-3)$ ;  $x^2 - 16 = (x-4)(x+4)$ ;  $x^2 - 3x = x(x-3)$ .  
*why:* Nothing can be cancelled until everything is in factor form.
- 2 Write the product:  $\frac{(x-3)(x+3)}{(x+4)(x-3)} \times \frac{(x-4)(x+4)}{x(x-3)}$ .  
*why:* In a product of fractions any top factor can cancel with any bottom factor.
- 3 Cancel: the  $(x+4)$  pair, and one  $(x-3)$  from the first numerator with the first denominator.  
*why:* There are two  $(x-3)$ s on the bottom but only one on top, so one survives below.
- 4 What remains:  $\frac{(x+3)(x-4)}{x(x-3)}$ .  
*why:* Count the leftovers carefully - this is where marks are lost.

#### Where it goes wrong

- Cancelling both  $(x-3)$  factors and losing one from the denominator.
- Cancelling terms rather than factors, e.g. striking out the  $x^2$ s.

**Takeaway** Factorise fully, then cancel factor-by-factor and count how many copies of each remain.

**Question 3** ALG-08 · GCSE algebra · spec M4.19

The  $n$ th term of a sequence is  $\frac{n^2 + 1}{2n}$ . Which of the following statements is/are true?

- I The 5th term is 2.6.
- II Every term after the first is greater than the previous term.
- III Some term of the sequence equals 10.

**Answer: E**

I:  $\frac{26}{10} = 2.6$ . II:  $u_n = \frac{n}{2} + \frac{1}{2n}$ ;  $u_1 = 1$ ,  $u_2 = 1.25$ , and for  $n \geq 1$  the increase  $\frac{1}{2} - \frac{1}{2n(n+1)} > 0$ , so increasing. III:  $n^2 - 20n + 1 = 0$  has no integer solutions (discriminant 396 is not a square), so false.

### Full method - how and why

Three claims about a sequence. Splitting the formula makes the behaviour transparent.

1 I:  $u_5 = \frac{25 + 1}{10} = \frac{26}{10} = 2.6$ . **True.**

why: Direct substitution.

2 Split the formula:  $u_n = \frac{n^2 + 1}{2n} = \frac{n}{2} + \frac{1}{2n}$ .

why: Splitting shows one growing part and one shrinking part - much easier to reason about.

3 II:  $u_{n+1} - u_n = \frac{1}{2} - \frac{1}{2n(n+1)}$ , which is positive for all  $n \geq 1$  since  $\frac{1}{2n(n+1)} \leq \frac{1}{4}$ . **True.**

why: The  $\frac{n}{2}$  term grows by  $\frac{1}{2}$  each step, while the  $\frac{1}{2n}$  term shrinks by less than that.

4 III: solve  $\frac{n^2 + 1}{2n} = 10 \Rightarrow n^2 - 20n + 1 = 0$ , whose discriminant  $400 - 4 = 396$  is not a perfect square, so  $n$  is not an integer. **False.**

why: Terms exist only for whole-number  $n$ , so a non-integer solution means no term equals 10.

5 I and II only.

why: Match to the options.

### Where it goes wrong

- Accepting III because the equation has real solutions - they must be **positive integers**.
- Testing II on only the first two terms.

**Takeaway** Split a rational formula into simpler pieces; and remember sequence indices must be whole numbers.